Course
Game Design: Theory and Practice

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5th Year Computer Science Engineering - Software Engineering

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Chapter 1
Game Design & Development

Chapter Plan

- Introduction
- Game Designer: Definition and Tasks
- Game Testers
- Gameplay
- Game Designer Role
- Gamers Expectations: What Players Want
- Brainstorming Ideas: Where do Ideas come from ?
- Game Design Document
- Game Design vs Game Development
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Introduction

Game design is the process of planning and creating how a video game works, what players do, and what makes the game fun, challenging, and enjoyable.

Think of it as designing the rules, gameplay, challenges, and experience of a game before and during its development.

It is the game's design that dictates the form and shape of the game's gameplay, and this is the factor that differentiates artistic medium from all others.

Game design is a critical field of the game industry because it is responsible for the designed game failure or success.

Game Designer

The game designer is the person who designs the game, who thereby establishes the shape and nature of the gameplay. They make decisions about:

- **Gameplay:**
  What can the player do? Jump, fight, build, explore, or solve puzzles?

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- **Gameplay**
- **Rules and mechanics:**
  How does the game work? For example, how does a player earn points or lose health?

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- **Gameplay**
- **Rules and mechanics**
- **Levels and maps:**
  Where does the game take place? What obstacles and secrets are hidden in the world?

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- **Gameplay**
- **Rules and mechanics**
- **Levels and maps**
- **Characters and story:**
  Who are the characters, what are their goals, and what happens in the game?

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- **Gameplay**
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- **Levels and maps**
- **Characters and story**
- **Progression and difficulty:**
  How does the game become more challenging as the player advances?

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- **Levels and maps**
- **Characters and story**
- **Progression and difficulty**
- **User experience:**
  Are the controls intuitive? Is the game fun and understandable?

Example: Minecraft

1. **Define the Gameplay:**
   Players collect wood, mine resources, craft tools, and build houses.
2. **Create the rules:**
   Players' health is affected by mobs, falling also affects the health but falling into lava is deadly.
3. **Design the World:**
   Choose the biomes, terrain, caves, villages, and places players can explore.







Game Testing

Or playtesting, is the process of evaluating a game by having people play it to assess its fun, gameplay mechanics, difficulty, and controls. Unlike debugging, which focuses on fixing technical errors, playtesting aims to improve the player experience and make the game more enjoyable.

Playtesters should provide specific and constructive feedback to identify gameplay problems and suggest improvements. A good playtesting process involves different types of players, including experienced gamers and beginners, to obtain diverse and objective opinions.

Gameplay

The gameplay is the component of computer games that is found in no other art form: interactivity. A game's gameplay is the degree and nature of the interactivity that the game includes, i.e., how players are able to interact with the game-world and how that game-world reacts to the choices players make.

It refers to the rules, controls, difficulty, and interactions that shape the player's experience. A key design principle is the 3Cs rule: **Camera, Controls, and Character,** which highlights three interconnected elements of the game system. Another fundamental concept is the gameplay loop, a recurring cycle in which the player pursues a goal, faces a challenge, and receives a reward.

Game Designer's Role

The game designer role in build the game is to understands what **motivates** players to continue and finish the game. The game designer has to keep players **interested** .

Motivation	Meaning
Overcoming a challenge	The player wants to solve a difficult problem or defeat an opponent.
Following a story	The player wants to discover what happens next.
Discovering new mechanics	The player enjoys learning new abilities or ways to play.
Expecting a reward	The player always expects to win something in return.

Game Designer's Role

Examples of possible rewards:

- **Feedback:** Visual or sound effects that make an action satisfying.
- **Collecting an object:** Obtaining an item that the player can keep or collect.
- **Feeling powerful:** Gaining a bonus or ability that makes the character stronger.
- **Sense of accomplishment:** Satisfaction after completing a goal or a difficult task.
- **Experiencing an emotion:** Feeling excitement, speed, fear, or adrenaline.
- **Competition or cooperation:** Satisfaction of beating an opponent or helping another player.

Game Designer's Role

The designer must understand these motivations and incorporate them into the game to make it engaging.

For example, in a platform game, the designer might create:

- Difficult levels for players who enjoy challenges.
- A story that unfolds as players progress.
- New abilities that players unlock.
- Rewards for completing levels.
- Sound and visual effects that make actions satisfying.

The key idea: a game designer must understand what motivates players and use appropriate challenges, progression, and rewards to encourage them to keep playing.

Example: The Last of Us: Part I

The designer carefully combines gameplay mechanics and storytelling to create challenges, build emotional connections, and make players feel invested in the characters. For example, switching from Joel to Ellie is not just a change of character: it changes the gameplay experience and helps players develop empathy for Ellie.

Bruce Straley, Game Director, Artist and Designer
"Crafting Emotional Beats in The Last of Us,"

Example: The Last of Us: Part I

- **Sense of accomplishment:**
  Surviving dangerous encounters gives players a sense of achievement.
- **Challenge the player's expectations:**
  Joel is gravely injured. His movement becomes difficult, and even sprinting causes him to collapse.The player is accustomed to controlling a powerful survivor, but suddenly struggles to keep him alive.
- **Emotional reward** :
  The player takes control of Ellie. She is smaller, more vulnerable, and must face dangers without Joel's usual strength. Playing as Ellie helps players understand her fears and experiences, strengthening their emotional connection to her.

Gamers Needs: What Players Want

The first question we should consider is: why do players play games? Why do they choose to turn on their computer or console and run Halo instead of visiting the art museum or going to see a movie? What is unique about computer games versus other human entertainment media? What do games offer that other activities do not?

Gamers Needs: What Players Want

- Players want to socialize
- Players want to be challenged
- Players want a dynamic experience even when playing solo games
- Players want to explore
- Players want to fantasize and live their dreams
- Players want to interact

Gamers Needs: What Players Expect

- Coherence and consistency
- Clear rules and understandable game-world's bounds
- Freedom
- Progression and to accomplish tasks incrementally
- Immersion
- Fair Chances and balanced difficulty
- Active participation
- Reasonable solutions to work
- Not hopelessly get stuck

Keep in mind !
Players do not know what they want but they know when it is missing

Brainstorming a Game Idea: Where do ideas come from ?

Brainstorming is a collaborative or individual technique for generating a large number of ideas to solve a problem or design a game feature.

Brainstorming rules:

1. Defer Judgment
2. Aim for quantity
3. Encourage unusual ideas
4. Build on their ideas

Brainstorming helps game designers explore many possible solutions, but an idea should ultimately be selected based on how well it supports the intended player experience, the game's objectives, and its practical constraints.

Game Design Document

GDD = Game Design Document (Document de conception de jeu vidéo).

A GDD describes the game's concept, rules, mechanics, content, interactions, and progression. It serves as a reference for the people involved in developing the game.

Although there are no rules for GDD structure, it typically follows the example below:

- Table of Contents
- Introduction/Overview
- Game Mechanics
- Artificial Intelligence (Game Intelligence)
- Game Elements
- Story Overview
- Game Progression
- System Menus

Game Design Document

1. **Introduction/Overview:**
   - Title and genre.
   - Target audience and platform.
   - Main concept and objectives.
   - What makes the game original.
2. **Game Mechanics:**
   - Movement, jumping, attacking, collecting objects.
   - Rules for winning and losing.
   - Resources, points, lives, and limitations.
   - Conditions for completing an objective.

Game Design Document

3. **Game Intelligence:**
   - Enemy behaviour and reactions.
   - Non-player character (NPC) behaviour.
   - Dynamic difficulty or environmental responses.
   - Feedback after an action.

4. **Game Elements:**
   - Characters and enemies.
   - Objects, weapons, and resources.
   - Environments, levels, and obstacles.
   - Sound, music, animations, and visual assets.
   - Interface elements.

Game Design Document

5. **Story Overview:**
   - Setting and context.
   - Main characters and their motivations.
   - Main conflict.
   - Important events and possible endings.

6. **Game Progression:**
   - Levels, missions, and objectives.
   - Increasing difficulty.
   - New abilities and rewards.
   - Unlocking areas or content.
   - Conditions for moving to the next stage.

Game Design Document

7. **System Menus:**
   - Main menu: start a game, load a game, or quit.
   - Pause menu: resume, restart, or return to the main menu.
   - Save and load systems.
   - Settings: sound, graphics, controls, and language.
   - Confirmation messages and navigation between menus.

Game Design vs Game Development

Game Design	Game Development
Plans the rules and mechanics	Implements them in code
Designs levels and challenges	Builds those levels in the game engine
Defines player goals and progression	Creates the systems that support progression
Focuses on the player’s experience	Includes programming, art, sound, tests…

Game Development Tools

Game development tools are software applications and technologies used to design, create, program, test, debug, and publish video games. They help developers build game environments, implement mechanics, create visual and audio assets, and manage the development process. To develop a game we need:

- Game engines
- Programming tools (IDEs)
- 2D/3D modeling tools
- Graphic design tools
- Audio Tools
- Version control tools
- Testing and profiling tools

References

- Game Design: Theory and Practice Wordware Game Developer's Library 2nd Edition, 2005 by Richard Rouse.
- Game Design Workshop: A Playcentric Approach to Creating Innovative Games 5th Edition, 2024 by Tracy Fullerton.